

CUET PG MCA (OPERATING SYSTEM PYQS)

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Topics Covered: CPU Scheduling, Process Management, Deadlock, Synchronization, Memory Management (Paging, Segmentation), Virtual Memory, Page Replacement, Disk Scheduling, Interprocess Communication.

Question 1

CUET PG 2021 | CPU Scheduling

Which of the following scheduling algorithms gives minimum average waiting time?

- (a) shortest job first
- (b) Round robin
- (c) priority
- (d) First come first serve

Question 2

CUET PG 2021 | CPU Bound

A CPU-bound program might have

- (a) a few long CPU bursts
- (b) a few short CPU bursts
- (c) many short CPU bursts
- (d) None of the above

Question 3

CUET PG 2023 | Disk Scheduling

The amount of time required to read a block of data from a disk into memory is composed of seek time, rotational latency and transfer time. Rotational latency refers to

- (a) the time it takes for the platter to make a full rotation
- (b) the time it takes for the read-write head to move into position over the appropriate track
- (c) the time it takes for the platter to rotate the correct sector under the head
- (d) to reduce number of bits in the field of instruction

Question 4

CUET PG 2024 | Process Creation (fork)

How many child processes will be created by following fork() system call?
fork(); fork(); fork(); fork(); fork();

- (a) 4
- (b) 16
- (c) 15
- (d) 3

Question 5

CUET PG 2024 | Banker's Algorithm

The current allocation and maximum requirement of different types of resources for four processes are given. Consider the following four statements:

- (A) $P2 \rightarrow P4 \rightarrow P1 \rightarrow P3$ is a safe sequence
- (B) $P4 \rightarrow P2 \rightarrow P1 \rightarrow P3$ is a safe sequence
- (C) $P4 \rightarrow P2 \rightarrow P3 \rightarrow P1$ is a safe sequence
- (D) $P1 \rightarrow P4 \rightarrow P2 \rightarrow P3$ is a safe sequence

Identify correct statements from the given options.

- (a) A, B and D only
- (b) A, B and C only
- (c) B, C and D only
- (d) A, B, C & D

Question 6

CUET PG 2024 | Deadlock (Min Resources)

An operating system contains 4 user processes each requiring 5 units of resource R. The minimum number of required units of R such that no deadlock will ever occur is

- (a) 20
- (b) 4
- (c) 17
- (d) 15

Question 7

CUET PG 2024 | Deadlock & Synchronization

Match List-I with List-II

List-I: (A) Critical Region (B) Working Set (C) Deadlock (D) Wait/signal

List-II: (I) Circular wait (II) Condition variable (III) Principle of locality (IV) Mutual exclusion

- (a) A-IV, B-III, C-I, D-II
- (b) A-IV, B-III, C-II, D-I
- (c) A-III, B-IV, C-II, D-I
- (d) A-III, B-IV, C-I, D-II

Question 8

CUET PG 2024 | Paging (EAT)

In a computer if the page fault service time is 10 ms and average memory access time is 30 ns. If one page fault is generated for every 10^6 memory accesses. What is the effective access time for the memory?

- (a) 21 ns approximate
- (b) 25 ns approximate
- (c) 30 ns approximate
- (d) 40 ns approximate

Question 9

CUET PG 2024 | FIFO & Belady's Anomaly

How does the number of page frames affect the number of page faults for a given memory access pattern in FIFO page replacement algorithm?

- (a) Increasing the number of page frames always decreases the number of page faults.
- (b) Increasing the number of page frames may increase or decrease the number of page faults depending on the memory access pattern.
- (c) Increasing the number of page frames always increases the number of page faults.
- (d) Increasing the number of page frames has no effect on the number of page faults.

Question 10

CUET PG 2024 | Segmentation

On a system using simple segmentation, following is the segment table:

Segment 0: Limit 500, Base 1000

Segment 1: Limit 200, Base 2000

Segment 2: Limit 300, Base 2500

Segment 3: Limit 100, Base 1700

What is the physical address for the logical address 2, 212?

- | | |
|----------|----------|
| (a) 2712 | (b) 512 |
| (c) 2212 | (d) 2800 |

Question 11

CUET PG 2024 | Virtual Memory

Decreasing the RAM of a computer typically leads to which of the following outcomes?

- | | |
|------------------------------|-------------------------------|
| (a) Virtual memory increases | (b) Page faults increases |
| (c) Page faults decreases | (d) Segmentation faults occur |

Question 12

CUET PG 2024 | Virtual Address Space

Consider a system with 1K pages and 512 frames and each page is of size 2KB. How many bits are required to represent the virtual address space memory.

- | | |
|-------------|-------------|
| (a) 20 bits | (b) 21 bits |
| (c) 11 bits | (d) 16 bits |

Question 13

CUET PG 2025 | OS Layers

Arrange the following layers of MS DOS operating system starting from inner most to outer most:

(A) ROM BIOS Device drivers

(B) Resident System Program

(C) MS DOS Device Drivers

(D) Application Program

- | | |
|------------------------|------------------------|
| (a) (A), (C), (B), (D) | (b) (D), (B), (C), (A) |
| (c) (B), (A), (D), (C) | (d) (C), (B), (D), (A) |

Question 14

CUET PG 2025 | CPU Scheduling

Which of the following scheduler/schedulers is/are also called CPU scheduler?

(A) Short Term Scheduler

(B) Long Term Scheduler

(C) Medium Term Scheduler

(D) Asymmetric Scheduler

- | | |
|---------------------------|---------------------------|
| (a) (A), (B) and (D) only | (b) (A), (B) and (C) only |
| (c) (A), (B), (C) and (D) | (d) (A) only |

Question 15

CUET PG 2025 | Page Replacement (LRU)

Consider a page reference string as: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2. Assume that there are 3-page frames available. Calculate the total number of page faults for the above reference string if LRU policy is used for page replacement.

- | | |
|--------|--------|
| (a) 10 | (b) 8 |
| (c) 9 | (d) 11 |

Question 16

CUET PG 2025 | Deadlock

A situation where two or more processes are blocked, waiting for resources held by each other is called:

- (a) Pooling
- (b) Deadlock
- (c) Thrashing
- (d) Paging

Question 17

CUET PG 2025 | Deadlock Recovery

Which of the following is a way to recover from the deadlock which has already occurred?

- (a) Process termination
- (b) Non preemption of resources
- (c) Banker's algorithm
- (d) Circular wait

Question 18

CUET PG 2025 | Deadlock Conditions

Complete the following statement by choosing the correct option.

For a deadlock to occur, the four conditions namely Mutual Exclusion, Hold and wait, No preemption, circular wait _____

- (a) May or may not hold
- (b) The circular wait does not imply hold and wait condition
- (c) Are completely independent
- (d) Must hold simultaneously

Question 19

CUET PG 2025 | CPU Scheduling Match

Match List-I with List-II

List-I (CPU Scheduling Algorithm): (A) FCFS (B) Round Robin (C) Multi level queue scheduling (D) Multi level Feedback Queue

List-II (Feature): (I) FCFS + preemption (II) Allows the processes to move between queues (III) Often long average waiting time (IV) Permanent assignment of processes to one specific queue

- (a) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
- (b) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
- (c) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
- (d) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Question 20

CUET PG 2025 | Disk Scheduling

Which disk scheduling algorithm looks for the track closest to the current head position?

- (a) LOOK
- (b) SSTF
- (c) FCFS
- (d) SCAN

Question 21

CUET PG 2025 | Disk Scheduling Match

Match List-I with List-II

List-I: (A) Seek Time (B) Access Time (C) Rotational Latency (D) Disk Bandwidth

List-II: (I) Total number of bytes transferred, divided by the total time between the first request for service and the completion of the last transfer (II) Time taken for the disk to rotate the desired sector to the disk head (III) Seek Time + Rotational Latency (IV) Time taken for the disk arm to move the heads to the cylinder containing the desired sector

- (a) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
- (b) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
- (c) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
- (d) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Question 22

CUET PG 2025 | Scheduling

Which CPU scheduling algorithm prefers the process with the shortest burst time?

- (a) FCFS
- (b) SJF
- (c) Round Robin
- (d) Priority Scheduling

Question 23

CUET PG 2025 | Process Section

Consider a typical process P in the critical section. Arrange the following statements of code to make a valid general structure.

(A) Critical section (B) Remainder section (C) Entry section (D) Exit section

- (a) (C), (A), (D)
- (b) (C), (A), (B), (D)
- (c) (C), (A), (D), (B)
- (d) (C), (B), (D), (A)

Question 24

CUET PG 2025 | Memory Hierarchy

Consider the following type of memories.

(A) Hard Disk Drive (HDD) (B) Cache Memory (C) Random Access Memory (D) Register

Arrange the above memories according to their access speed (from fastest to slowest)

- (a) (D), (C), (B), (A)
- (b) (A), (C), (B), (D)
- (c) (B), (D), (C), (A)
- (d) (D), (B), (C), (A)

Question 25

CUET PG 2025 | Synchronization

The Dining Philosopher' problem can be solved by:

- (a) Use of semaphores
- (b) Use of overlays
- (c) Mutual exclusion
- (d) Bounded waiting

Question 26

CUET PG 2025 | Paging

Details of a paging system for memory management are as follows:

Logical address space: 32 KB

Page Size: 4 B

Physical Memory size: 64 KB

The number of pages in the logical address space and number of page frames in physical memory, respectively are:

- (a) 4, 16
- (b) 4, 12
- (c) 8, 16
- (d) 3, 16

Question 27

CUET PG 2025 | Memory Management

External fragmentation occurs

- (a) When enough total memory space exists to satisfy a request, but it is not contiguous, storage is fragmented into a large number of small holes.
- (b) When enough total memory space exists to satisfy a request, which is contiguous, storage is fragmented into a large number of small holes.
- (c) When memory is empty.
- (d) Always.

Question 28

CUET PG 2026 | Interrupt / System Call

The occurrence of an event is usually signaled by an interrupt from software by executing a special operation, which is called as _____:

- | | |
|---------------------|------------------------|
| (1) System call | (2) Bootstrap Program |
| (3) Service Routine | (4) Interrupt - Vector |

Question 29

CUET PG 2026 | Interprocess Communication

Cooperating process require an Interprocess communication mechanism. This works on which of the following fundamental models?

(A) Shared Memory (B) Cascading Termination (C) Message Passing (D) Cooperating

- | | |
|------------------|------------------|
| (1) C and D only | (2) A and C only |
| (3) A and D only | (4) B and D only |

Question 30

CUET PG 2026 | CPU Scheduling Match

Match List-I with List-II (Scheduling Description).

List-I (Scheduling): (A) First-Come, First Served (B) Shortest-Job-First (C) Priority Scheduling (D) Round Robin

List-II (Description): (I) CPU is allocated to the process with the highest priority (II) The Process that requests the CPU first is allocated first (III) Algorithm associates with each process the length of next CPU burst (IV) A pre-emption is added to switch between processes

- | | |
|----------------------------|----------------------------|
| (1) A-II, B-III, C-I, D-IV | (2) A-I, B-II, C-III, D-IV |
| (3) A-II, B-I, C-IV, D-III | (4) A-III, B-II, C-I, D-IV |

Question 31

CUET PG 2026 | Memory Management (Paging)

_____ is a memory management scheme that permits the physical address space of a process to be non-contiguous.

- | | |
|---------------------|---------------------|
| (1) Dynamic loading | (2) Address binding |
| (3) Paging | (4) Partitioning |

Question 32

CUET PG 2026 | Critical Section

Which of the following requirement must be satisfied to solve a critical section problem?

(A) Mutual Exclusion (B) Race Condition (C) Progress (D) Bounded waiting

- | | |
|---------------------|---------------------|
| (1) A, B and D only | (2) A, C and D only |
| (3) B, C and D only | (4) A, B and C only |

Question 33

CUET PG 2026 | Bounded Buffer (Semaphores)

Arrange the following semaphore operations in the correct sequence for the producer process in the Bounded-Buffer Problem using semaphores empty, full, and mutex:

(A) Signal (full) → indicating that the new item has been produced and added to the buffer

(B) Wait (empty) → Checks if there is space in the buffer to add new item

(C) Produce an item and store it in the buffer

(D) Wait (mutex) → Enters the critical section to insert the item safely

(E) Signal (mutex) → Leaves the critical section after inserting the item

- | | |
|-------------------|-------------------|
| (1) A, E, C, B, D | (2) C, B, A, D, E |
| (3) D, E, A, C, B | (4) B, D, C, E, A |

Question 34

CUET PG 2026 | Deadlock

Which state occurs when two or more processes are waiting indefinitely for an event that can be caused only by one of the waiting processes.

- (1) Deadlock state
- (2) Synchronisation state
- (3) Resource allocation state
- (4) Preemption state

Question 35

CUET PG 2026 | Deadlock Assertion

Assertion (A): Deadlocks occur in a multi programming environment when each thread in a set is waiting for an event that can be caused only by another thread in the same set.

Reason (R): A thread must always request a resource before using it and release it after using it. If requested resources are not available the thread enters a waiting state until the resources becomes free.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

Question 36

CUET PG 2026 | Page Fault Handling

Sequence the following to compute the effective access time needed to service a page fault.

- (A) Receive an interrupt from the disk I/O subsystem
- (B) Trap the operating system and save the user registers and process state
- (C) Determine that the interrupt was a page fault
- (D) Correct the page table and other tables to show that the desired page is now in memory
- (E) Restore the user registers, process state and new page table, and then resume the interrupted instruction

- (1) A, B, C, D and E
- (2) B, C, D, E and A
- (3) A, D, B, C and E
- (4) B, C, A, D and E

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ANSWER KEY (2021 – 2026)

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	a	2	a	3	c	4	c	5	b
6	c	7	a	8	d	9	a	10	a
11	b	12	b	13	a	14	d	15	b
16	b	17	a	18	d	19	b	20	b
21	d	22	b	23	c	24	d	25	a
26	a	27	a						

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